

Masai IITr Capstone Project

Q1: Capstone Project and Title:

LedgerLens – **OnRecord**: Reads multiple photographed receipts and extracts invoice information including vendor, date, items, total prices with confidence scores and a human-review queue for low confidence scores.

Q2: Code Repository Links:

GitHub: <https://github.com/devanshrs45/ledgerlens>

Q3: Deployed App URL + Test Credentials:

URL: <https://on-695cd5319e554ecd822f62de2375398a.ecs.ap-south-1.on.aws/>

Test Credentials: N/A

Q4: Demo Video Link:

YouTube (Private): w

Q5: Build Note and Open-Ended Stretches:

SHIPPED:

Pipeline:

Photo Upload/Batch photos upload (Upto 25) →

→ OpenAI Moderation (Threshold: 0.5) →

→ Groq Vision Extraction with per-field confidence scores (Threshold: 0.75) →

→ Pydantic validation for schema →

→ Flags fields based on confidence scores below 0.75 →

→ Flags fields based on Deterministic arithmetic checks on values →

→ Watermarks Image and Stores. PII Redaction and logging →

→ Human review queue with option to edit field values (both flagged and others) →

→ Approve/Reject Reviewed Receipt →

→ Collective records of receipts with its extracted details.

Deployed on AWS using S3, RDS, ECS, ECR.

Certain specifications in pipeline:

- Single/Multiple photos can be uploaded. Image type validation
- Moderation can be toggled true/false
- Moderation runs before Vision Extraction: If Image is blocked, it is not extracted by Groq, reducing unnecessary token utilisation. If Allowed, image resized and extracted.
- Every extracted field has a {value, confidence}
- Fields flagged if score is below 0.75 and sent to review. Deterministic flags check arithmetic of the values (Such as sum of amounts = total) and flag if not making sense.
- Adds Visible watermark and stores in S3. PII Redaction and logging
- Human review can edit flagged fields and have an option to edit other fields too.
- Observes metrics like moderation/extraction latency, total cost (with each document cost), number of documents processed, approved/rejected, etc.
- Deployment on AWS: S3, RDS, ECR, ECS

KEY DECISIONS:

LLM, Tools, Services, Language:

- Groq with Qwen Vision Model for vision extraction (Free)
- OpenAI for Moderation (Moderation is free, but not on free-tier account)
- JSON Structure with Pydantic
- Vanilla HTML/CSS/JS was used for Frontend
- AWS for hosting and storage (Used \$200 credits given in free tier, No extra cost)
 - S3 Stores the Watermarked Images
 - RDS stores the records (PostgreSQL)
 - ECR Stores Docker Image
 - ECS Express Mode runs the application
 - Container built in AWS Cloudshell

Secondary/Feature choices:

- Batch upload upto 25 files.
- Added 'Discount' and 'Additional_Charges' in invoice schema. Discount measures any discount, coupons, sale, off, etc. Additional_Charges measures cost like delivery fee, tips, environmental fee, etc.
- Date extraction ambiguity reduced: when day ≤ 12 , and date format is DD/MM/YYYY or MM/DD/YYYY (as different places use different formats), model checks location and other factors instead of guessing the month/date.
- Some Indian specific PII was redacted. Aadhar was redacted through Verhoeff Algorithm to avowing redacting Invoice Numbers instead of aadhar.
- Deterministic checks based on values extracted. For example, checks if (Sum of Items' Costs = Subtotal) or (Subtotal + Tax + Additional – Discount = Total), etc. This also rechecks if the extracted values are correct and points out any discrepancy in the receipts or extraction values.
- Human review can edit flagged field values or other values too.

STRETCHES:

Working:

- Batch processing (static/app.js)
- Day date/month ambiguity reduced (not completely eliminated)
- Discount and Additional charges added
- Deterministic Arithmetic rule-based flagging (app/extraction.py)
- Aadhar redaction using Verhoeff pattern (app/pii.py)
- Some Indian PII patterns like GSTIN, PAN (app/pii.py)
- Reject after review (app/routers/review.py)
- Non-flagged values are also editable when in review queue (static/app.js)
- Export extracted batch records to CSV

Partial:

- Line Items: Flags when not confident or arithmetic does not make sense but cannot edit it for each item specifically (like qty, price).
- Phone Number PII redaction (after extraction): It redacts phone number correctly, but may also redact invoice number if ≥ 10 digits.
- PII redaction is log only. Images stored in S3 or displayed in 'Records' show these PII values in image.

LIMITATIONS:

Model level:

- Groq free tier has limit of 8000 tokens per minute. Thus, I set a 4000 token cap on vision extraction (other tokens used for input and other things). Image resolution was also lowered, and might reduce accuracy on very small, blurry receipts.
- Batch processing will take time due to Groq limit, around 30 seconds per document.
- One document per receipt: each photo should contain a single receipt, otherwise it might be confused and extract with mixed results.
- Date day/month ambiguity is not completely resolved, the model cannot determine which date/month to keep, especially when location not given. Sometimes, vendors can store dates in other formats too not consistent with the location (example: Indian vendor storing receipt in MM/DD/YYYY form).
- A highly complex receipt with tables and lots of number may confuse the model to pick different numbers.

Extraction/Review/Storage/Deployment level:

- Each specific item's details (qty, price, etc) cannot be edited like invoice level details (vendor, total, etc). However, confidence scores and deterministic flags may flag these values if incorrect.
- In PII redaction, the phone number redaction me redact invoice numbers if ≥ 10 digits.
- PII redaction is log only. Images stored in S3 or displayed in 'Records' show these PII values in image.
- URL is public and no authentication. Anyone with the URL can upload, approve, reject and see previous document records (in Records Tab).
- Manual redeployment

FUTURE EXTENTION (Extra):

Main goal to reduce the above limitations

- Compute business metrics/values. Like, total earnings from all receipts, number of items sold, etc
- Full line-item editing: name, qty, price
- Sorting records based on parameters + Filters
- Reopen approved document to change if incorrect
- New Vision extraction model with higher complexity/rate-limit than Groq (In paid version)
- Redact PII in images
- User authentication on application (for each business, vender, etc)